LAB EXPERIMENT RECORD

Set 20: Stock Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
Name	Swetha S
Register Number	192424049
Date	13/10/2026
Day	Tuesday

AIM

To analyze the stock prices of three companies using R programming by creating a line chart, bar chart, and data table, and to combine these visualizations into an interactive Tableau dashboard.

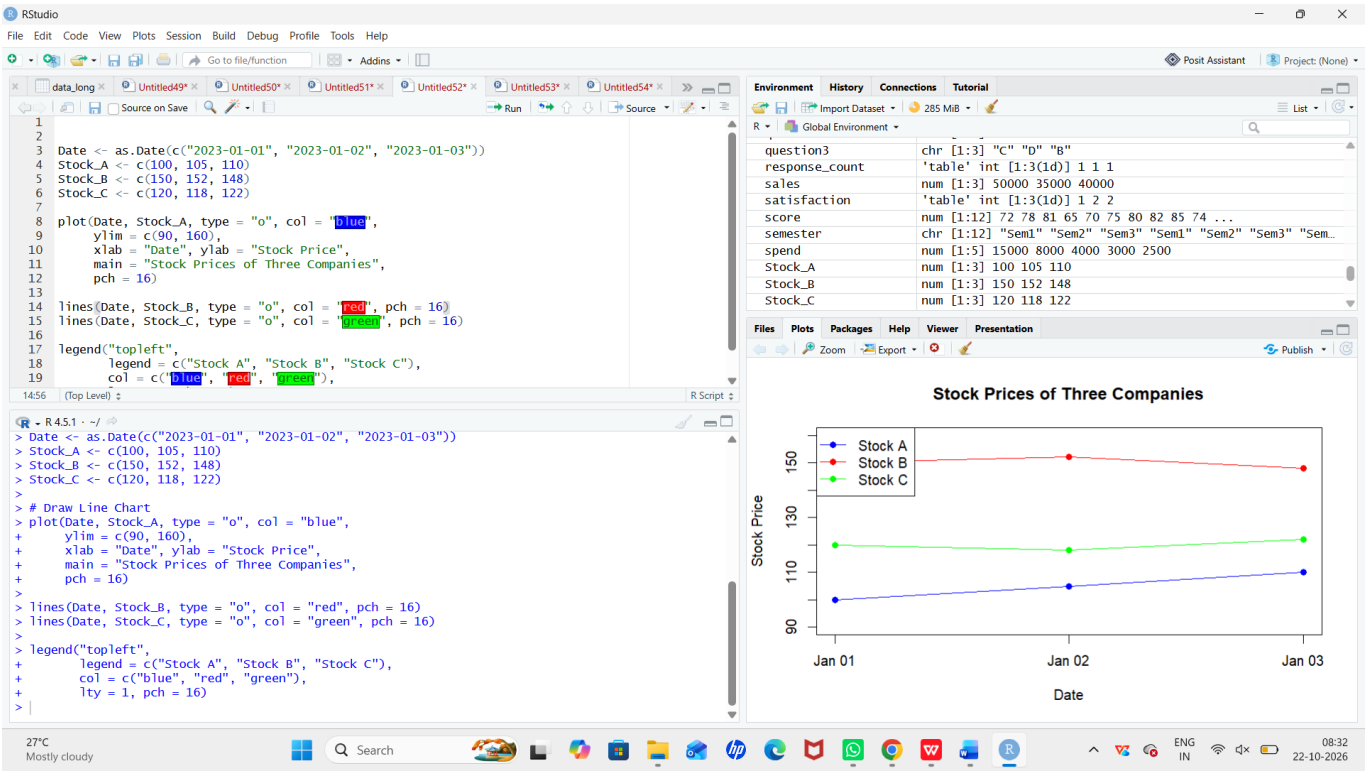
1. Create a line chart to visualize the stock prices of three companies Label the axes and title the chart.on

```
Date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03"))
Stock_A <- c(100, 105, 110)
Stock_B <- c(150, 152, 148)
Stock_C <- c(120, 118, 122)

plot(Date, Stock_A, type = "o", col = "blue",
     ylim = c(90, 160),
     xlab = "Date", ylab = "Stock Price",
     main = "Stock Prices of Three Companies",
     pch = 16)

lines(Date, Stock_B, type = "o", col = "red", pch = 16)
lines(Date, Stock_C, type = "o", col = "green", pch = 16)

legend("topleft",
       legend = c("Stock A", "Stock B", "Stock C"),
       col = c("blue", "red", "green"),
       lty = 1, pch = 16)
```



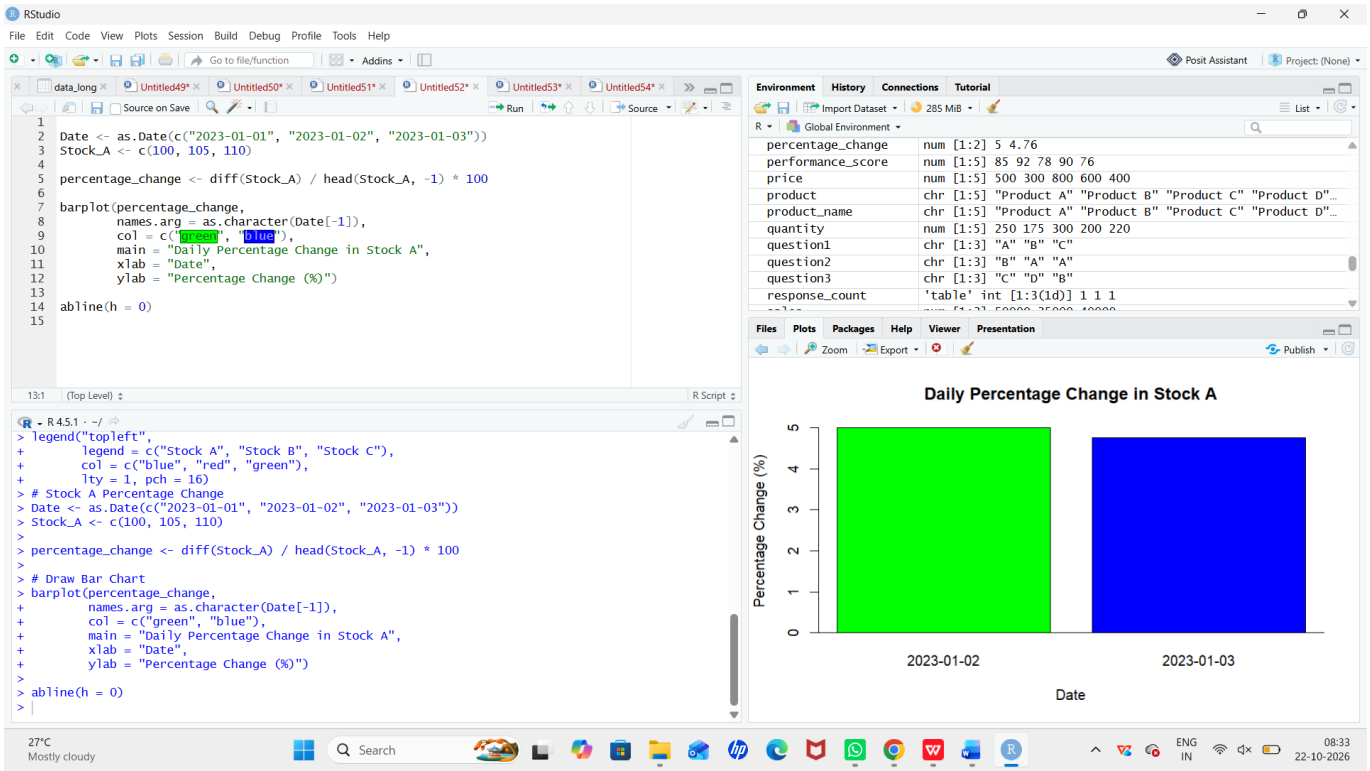
2. Generate a bar chart showing the daily percentage change in stock prices for Stock.

```
Date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03"))
Stock_A <- c(100, 105, 110)

percentage_change <- diff(Stock_A) / head(Stock_A, -1) * 100

barplot(percentage_change,
        names.arg = as.character(Date[-1]),
        col = c("green", "blue"),
        main = "Daily Percentage Change in Stock A",
        xlab = "Date",
        ylab = "Percentage Change (%)")

abline(h = 0)
```

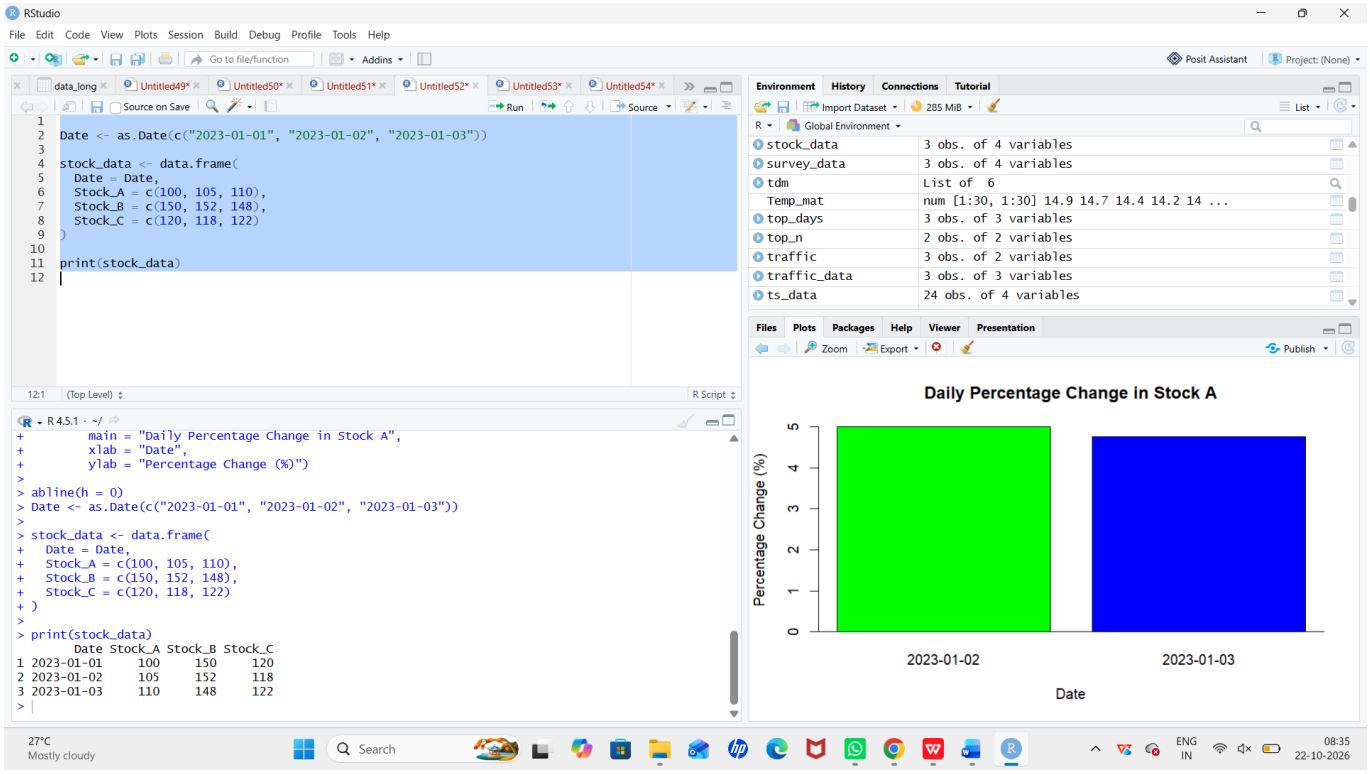


3. Build a table to display the stock price data for each company over the given period..

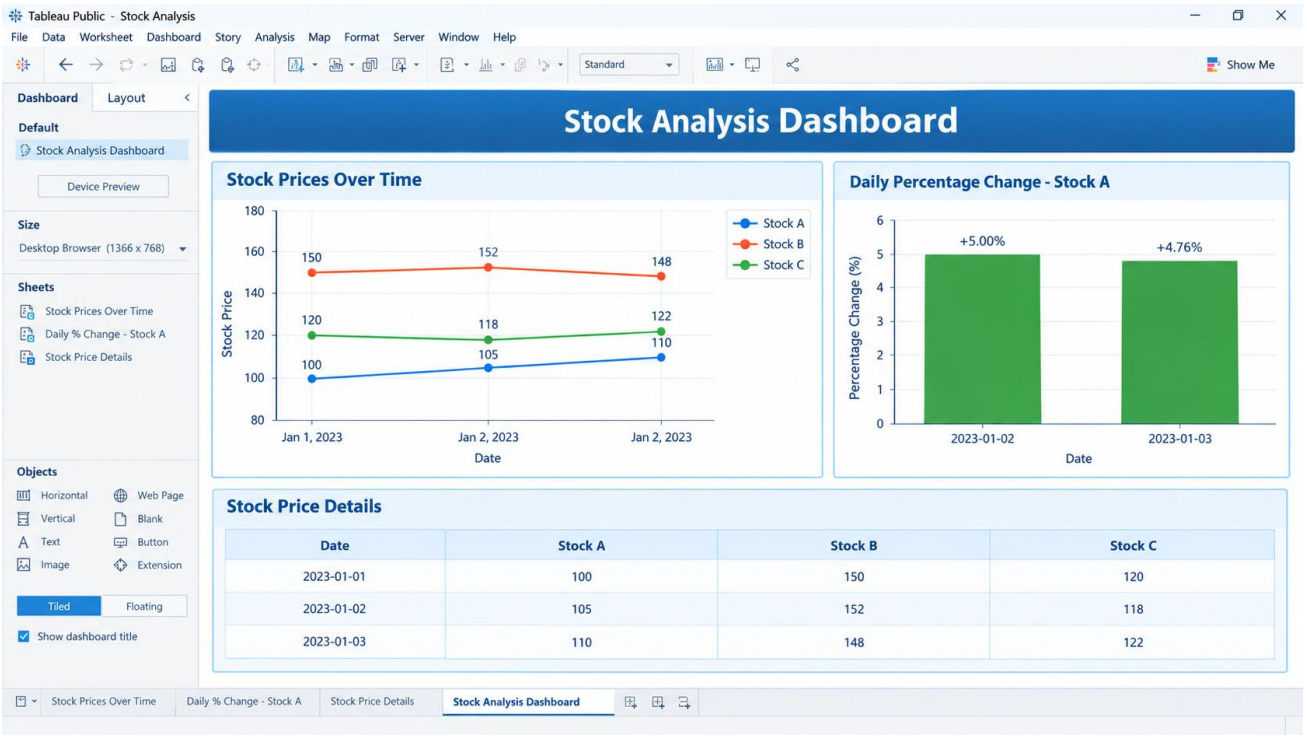
```
Date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03"))

stock_data <- data.frame(
  Date = Date,
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)

print(stock_data)
```



4.Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive



Procedure

1. Create a data frame containing the dates and stock prices of Stock A, Stock B, and Stock C.

2. Use a line chart to visualize stock price changes over time.

3. Calculate the daily percentage change in Stock A and represent it using a bar chart.

4. Display the stock price details in a table with proper labels.

5. Import the dataset into Tableau and create separate worksheets for the line chart, bar chart, and table.

6. Combine all three worksheets into a single interactive dashboard for stock price analysis.

Result

The stock prices of three companies were successfully analyzed using R programming. A line chart, a bar chart showing the daily percentage change in Stock A, and a stock price table were created. These visualizations were combined into an interactive Tableau dashboard for easy comparison and analysis of stock prices.